BREAKAGE OF WHEELS AND TIRES ON BRITISH RAILROADS.

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WE continue below the record of accidents on British railroads from breakages of wheels and tires, which was begun in the June number of the JOURNAL, and the purpose and object of which were there explained. The record then closed with the year 1864, and is now taken up from that point.

ACCIDENT REPORTS.

March 21, 1865, express train on Great Western Railway was derailed near Goring by the breaking of a tire under a passenger car. The train ran over 5 miles before the engine-driver saw that something was wrong and put on the brakes. The tire gave way at the weld, which was very defective, and opened out some 13 in. It was fastened in its place by a ring or circular key riveted down over the rim of the wheel center, but when the tire broke this ring came off altogether and dropped on the road-bed. The wheel center was broken to pieces by hammering over the ties and ballast.

November 12, 1866, a mixed train on the North British Railway was thrown from the track near Spittal by a broken tire under a freight wagon. All the passenger cars left the rails, and one of them was badly broken up. The broken tire was nearly new and was 2 in. thick on the tread; it was fastened to the wheel center by four ¾-in. rivets. At the place where it broke a bad flaw was apparent in the iron.

March 23, 1867, the locomotive of a passenger train on the Glasgow & Southwestern was thrown from the track near Kirkconnell by the breaking of a driving-wheel tire on the engine, while running about 40 miles an hour. The engine ran completely off the line and upset, injuring the fireman. The driving-wheels, a single pair, were 74 in. diameter; the tire was cast-steel, originally 2⅛ in. on tread, but reduced by wear and turning off to 1¼ in. It had been shrunk on the wheel and secured to it by six tapering, countersunk bolts ⅞ in. in diameter and having a shoulder about $\frac{1}{16}$ in. some ¾ in. from the face of the tire. The tire broke through one of the bolt-holes, and there was a flaw in the steel extending from the bolt-hole to the bend of the flange. The solid metal remaining was only about one-half the sectional area of the tire. The flaw extended to the surface and might have been seen in the form of a fine crack. The engine had been in the shop three months before and had the tires turned off. In this case, the first in this list of steel tires, the Inspector says: "I understand from the Locomotive Inspector, Mr. James Stirling, that cast-steel tires were introduced on this road in July, 1861, and since that time, including the present case, four altogether have broken. * * It is apparent, from what has happened on this line since cast-steel rolled tires were introduced in 1861, that some more efficient mode of fastening such tires to the wheels than has been hitherto adopted by this company is necessary for safety. There are many methods well known, which might be adopted, free from the very obvious defect, which is inseparable from the process, of materially weakening the tires by drilling bolt-holes through them, and I cannot think that any railway company is justified in fastening tires to wheels by such a process. If, however, it be contended that the drilling of such holes does not render the tire more liable to break when it has been, as in this instance, reduced to about one-half its original thickness, it is quite evident that a much larger number of bolts should be made use of to prevent tires from flying off the wheels when they do break under such circumstances as have been detailed respecting this accident."

November 27, 1868, a passenger train on the Great Eastern Railway was thrown from the track by the breaking of a tire on a trailing wheel of the engine. The tire broke into four pieces, respectively 1 ft. 4 in., 2 ft. 7 in., 7 ft. and 7 ft. 3 in. long. One of the broken pieces threw off a carriage, injuring three passengers. The tire was Krupp steel, had been running 5 years and had been turned down to about 1¼ in. on the tread. It was fastened on the center by hammering down the rim on wedge-shaped keys. In this case, one of the hammered portions of the rim gave way, allowing a key to fall out, and the tire to begin working; this was probably followed by the shifting of the tire. The Inspector does not consider this method of fastening tires a good one.

November 2, 1868, an express train on the Northeastern Railway was derailed near Hunslet by the breaking of the tire on a leading wheel of the engine. The tire broke at the weld and came off the wheel altogether; it was of wrought-iron, originally 2¼ in. thick, but reduced to 1⅝ in. by turning and wear. It had run 14,285 miles before being turned and 1,864 miles since. It was held to the wheel-center by six ⅞-in. tap-bolts, all of which broke off close to the inside of the tire. The weld was a very imperfect one. In this case the Inspector says: "The question naturally arises whether, so long as welded tires fastened only by bolts continue to be used for the wheels of railway rolling stock, some effectual mode of testing the soundness of the welds, particularly after the tires have been turned, ought not to be devised."

October 31, 1869, a passenger train on the Vale of Towy Railway was thrown from the track between Llanwrda and Glwnrhyd by the breaking of a tire on a leading wheel of the engine. The tire broke into five pieces, 3¼, 2¼, 3, 1¼ and 2¼ ft., respectively. The tire was fastened to the center by 6 bolts, and was dovetailed on the wheel also; it was of cast-steel. The probable cause of fracture was that the tire was made of too hard steel, and was put on the wheel too tight, that is, was shrunk too much in putting it on.

June 7, 1870, seven carriages of an excursion train on the Lancashire & Yorkshire road were thrown from the track near Hoghton, and one of them went over a bank, killing 2 passengers and injuring 27. The cause was a broken tire under a carriage. It was a new tire, in use only two months, made of Bessemer steel and 1¾ in. thick on tread; it was shrunk on the wheel and secured by 4 wrought-iron rivets, tapering from 1 to ¾ in. diameter. The rivet holes were drilled. Tests of the broken tire showed that the metal was very brittle.

December 26, 1870, passenger train on Great Northern line ran off the track at Marshmoor, owing to the fracture of a tire under a brake-van. The engine broke loose and went on, but 6 cars were derailed and 2 badly wrecked, killing 8 and injuring 3 persons. In this case the tire was steel, 42 in. diameter, nearly new, and was dovetailed into the center in front, and secured at the back by 8 clips or keys fitting into a recess in the tire. There were also 4 screws, ⅞ in. diameter, running through the rim and about half the thickness of the tire. It broke in 6 pieces. From evidence taken it seems that the tire was very hard; the weather at the time was cold, and the road-bed very hard.

January 3, 1871, passenger train on Midland Great Western (Ireland) was derailed near Oranmore by broken tire under a passenger carriage. The tire broke in two places, a piece 18 in. long being thrown completely off the wheel. No flaws were found. The tire was iron, and was fastened to the wheel by Cabry's patent, one edge of the tire being grooved and fitted on to the outside of the rim of the wheel-center, and the lap on the outside of the tire being hammered down over the inside edge of the rim.

January 5, 1871, passenger train on London & Northwestern was derailed at Holme station, wrecking several carriages, injuring the guard and 5 passengers. The accident was caused by the breaking of a tire under a brake-van. An extensive flaw was found in the weld (where the tire broke) which appeared at the surface as a slight crack. The tire was originally 2 in. thick, but had been worn and turned down to $1\frac{5}{16}$ in. A second break was through a bolt-hole. The Inspector's report says: "The tire was of iron of good quality, and fixed on the inner rim of a wrought-iron wheel having 9 spokes, by means of 5 conical-headed countersunk wrought bolts ¾ in. diameter, with screw-nuts on the inside of the inner rim or sole of the wheel-center. * * The London & Northwestern and the Caledonian companies have for many years used Mansell's wheels, and I am not aware that any

case has arisen in which a tire fastened on by this method has flown. * * This accident points out conclusively that wheels with tires fastened on as this one was should not be run at all on passenger trains. It is certain that the public traveling on railways in these carriages are subjected to increased risk from this cause in very severe weather; no matter whether the liability to fracture be due to the rigidity of the road-bed, to the greater strain on the tire by its contraction from frost, to the possibility of its being more brittle in frosty weather, or to combined action of all these causes. The increased risk is certainly incurred, and it is high time—10 years since this risk was first made clearly apparent during the winter of 1860–61, and the means of avoiding it clearly pointed out by the officers of the Board of Trade—that it should be incurred no longer."

January 5, 1871, passenger train on the Manchester, Sheffield & Lincolnshire line was derailed near Huntingdon by a broken tire under a passenger carriage. Four persons were hurt. The tire was of steel, shrunk on and fastened with four ¾-in. rivets; it was an old tire, and was worn down to 1⅝ .n. thick on tread. It broke into 6 pieces. There was a slight flaw perceptible at one of the breaks. This accident again shows, says the Inspector, the need of better methods of fastening tires to the wheels.

January 17, 1871, passenger train on Midland line was derailed near Draycott by a broken tire under a passenger carriage. The tire was of iron, fastened on by 4 bolts, tapering from 1⅛ to ¾ in. diameter. It broke into 3 pieces. This was another case of poor fastenings.

November 9, 1871, passenger train on Great Western line was derailed near Windsor by the breaking of a wheel under a passenger carriage. This was a Mansell wheel, 43 in. diameter. This wheel is made with a steel or iron tire and a cast-iron hub, the space between being filled by an annular disk of teak wood, made up of 16 pieces, 3½ in. thick and about 15 in. long. The hub has a circular flange on one side, and a heavy washer slips over the other end, the two being bolted together by 8 bolts, holding the wood blocks at that end. On the outer end the tire is made with a dovetail on each side and two circular iron rings, which fit into the dovetail and project considerably beyond the inside of the tire. These rings are fastened by 16 bolts, one passing through each section of the wood. In this case most of the wood broke from the center, leaving an irregular rim and the tire with no center. Only one of the bolts was broken. The wheel is to be improved in the future by increasing the size of the bolts and putting in a larger hub. The Inspector calls this a very good wheel.

November 8, 1871, passenger train on Midland line was derailed near Kingsbury Wood by a broken tire on a driving-wheel of the locomotive. This tire was of steel, 6 ft. 6 in. diameter, had run 80,110 miles, and had been turned and worn down from 2⅜ in. to 1⅜ in. thick on tread. It broke into two pieces, showing an old flaw at one of the breaks.

November 10, 1871, passenger train on Midland line was derailed at Borrowash by broken tire on a locomotive driving-wheel. This tire was of crucible steel, had run 87,800 miles and had been three times turned up; it was 1⅜ in. thick, having been reduced from an original 2⅜ in. It was shrunk on the wheel-center and secured to it by seven ⅞-in. screws tapped into the tire about 1 in. from the inside. The tire broke into three pieces and was badly distorted in shape. In this and the preceding case, the want of better fastenings for the tire is blamed for most of the damage done.

November 12, 1871, express train on Northeastern Railway was thrown from the track near Learnside by broken tire on a leading wheel of the locomotive. The whole train left the track and two carriages upset, injuring 3 passengers. The broken tire was nearly new; it was of cast-steel, was 54 in. in diameter and 2½ in. thick on tread. It had a flange on the outer face bearing against the rim of the wheel-center, and was fastened to the center by four ¾-in. countersunk bolts. It broke into four pieces, It appears that the steel was very hard, so that the workmen in the shop complained that all their tools were dulled in turning it up.

December 1, 1871, passenger train on the Northeastern line was derailed near Killingworth by a broken tire under a van or "horse-box," half of the train leaving the rails and injuring 2 trainmen and 7 passengers. The train ran nearly a mile before it stopped. The tire was of cast-steel rolled out, had been running over four years and had worn down from 2 to 1¾ in. on tread. It broke into 3 pieces, 1½, 3 and 5½ ft. long, respectively. None of the breaks were at bolt-holes, but one of them showed an old flaw extending over the larger part of the sectional area. The tire had a notch or lip projecting ¼ in. over the rim of the wheel in front, and was further fastened by four ¾-in. countersunk bolts. The Inspector says: "This accident adds another testimony to the extreme importance of securing tires to wheels by some method which shall prevent their leaving them in the event of fracture. This is more needful than ever, now that the use of tires of so comparatively uncertain a metal as steel is becoming general"

December 24, 1871, passenger train on Northeastern Railway was derailed near Aycliffe by a broken tire under a passenger car. The tire was of cast-steel, 48 in. diameter and was fastened on by the Beattie method, by which a projecting lip holds the tire to the wheel-center in front, while at the back there are 9 iron clips or keys, held in place by hammering down over them a projecting lip left on the tire for the purpose. The tire broke into 3 pieces, respectively, 34, 44 and 78 in. long. From marks found it was believed that the tire first broke several miles back from the point where it left the wheel altogether. The results on the permanent way were remarkable; 200 chairs, 8 ties and 1 steel rail (82 lbs. section) were broken and had to be replaced.

February 2, 1872, passenger train on Lancashire & Yorkhire road was derailed at Cooper Bridge by a broken tire under the tender. The tire was of i cnand riveted ton the wheel by four ⅞ in. rivets. The Inspector here agai takes occasion to recommend improved methods of fastening.

September 17, 1872, passenger train on Great Southern & Western (Ireland) was derailed near Mallow by a broken tire under a passenger carriage. Of this the Inspector says: "The tire that gave away was of cast-steel, made at Leeds in 1870. It made its first journey in August, 1870. It was fixed to the wrought-iron wheel center by five ⅞ in. screw-bolts, and had been shrunk on the wheel, the amount allowed for shrinkage being 0.05 in. in the diameter. The tire showed very little wear. The grain of the metal was good; but when tried in a hydralic press, the steel proved to be brittle. It broke, without yielding, under a pressure of 35 tons applied at the center of a 22 in. bearing. * * It is impossible to prevent tires from breaking. All those hitherto manufactured have done so occasionally. It is, however, possible to prevent their leaving the wheels when they do break."

October 16, 1872, passenger train on the Midland line was derailed near Dronley by broken tire under a passenger carriage, killing 1 passenger and injuring another. The tire was of iron, 39 in. diameter and 1¾ in. thick on tread: it was fastened to the wheel-center by 4 bolts. The tire broke in 4 pieces, 3 of the breaks being at bolt-holes.

December 6, 1872, passenger train on Midland Railway was derailed near Ambergate by broken tire under a passenger car. The tire was iron, fastened by 4 conical rivets, and broke through a rivet hole. Here improved methods of fastening are again recommended.

In the period covered by the part of the summary given above, steel tires appear in considerable number, although iron tires caused a majority of the accidents and were still much more largely in use than steel. With the steel tires the greater number of accidents seem to have been caused by the use of of too hard steel, lacking in the toughness needed in a tire.

As before, the accidents were almost entirely due to the breakage of tires. Only 2 cases of broken wheel-